

CO5_AT02_SHORT ANSWERS TEST

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CO5- AT2-Short Answer Test

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QUESTIONS

Q1 Does Floyd-Warshall compute single-source or all-pairs shortest paths? 2 marks

Q2 How many edges must a Minimum Spanning Tree of a graph with V vertices have? 2 marks

Q3 In a DAG, which node will always have an in-degree of zero in a topological ordering? 2 marks

Q4 Differentiate between directed and undirected graphs. 2 marks

Q5 Define a connected graph. 2 marks

5/5 answered

Q1 Does Floyd-Warshall compute single-source or all-pairs shortest paths? 2 marks

Does Floyd-Warshall compute single-source or all-pairs shortest paths?

Your Answer

Floyd-Warshall computes all-pairs shortest path. it finds the shortest paths between the every pair of vertices in the graph unlike algorithms such as dijkstras which compute single source shortest paths from one starting vertex to all others.

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Q2 How many edges must a Minimum Spanning Tree of a graph with V vertices have? 2 marks

How many edges must a Minimum Spanning Tree of a graph with V vertices have?

Your Answer

A Minimum Spanning Tree of a graph with V vertices must have exactly $V-1$ edges. this is because a spanning tree connects all V vertices without forming any cycles and the minimum number of edges required to connect V vertices in a tree structure is always $V-1$

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QUESTIONS

Q1

Does Floyd-Warshall compute single-source or all-pairs shortest paths?
2 marks

Q2

How many edges must a Minimum Spanning Tree of a graph with 5V5 vertices have?
2 marks

Q3

In a DAG, which node will always have an in-degree of zero in a topological ordering?
2 marks

Q4

Differentiate between directed and undirected graphs.
2 marks

Q5

Define a connected graph.
2 marks

5/5 answered

Q3 In a DAG, which node will always have an in-degree of zero in a topological ordering?

2 marks

In a DAG, which node will always have an in-degree of zero in a topological ordering?

Your Answer

In a DAG, the source node will always have an in-degree of zero in a topological ordering. This node appears first in the topological sort since it doesn't depend on any other node being processed before it.

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Differentiate between directed and undirected graphs.
2 marks

Q5

Define a connected graph.
2 marks

5/5 answered

Q4 Differentiate between directed and undirected graphs.

2 marks

Differentiate between directed and undirected graphs.

Your Answer

Directed graph : each edge has a specific direction, meaning it points from one vertex to another example: it represents as an ordered pair. This implies that if there is an edge from A to B, it does not necessarily mean there is an edge from B to A.
Undirected graph : edges have no direction. They simply represent a connection between two vertices example: represented as an unordered pair. meaning if A is connected to B, then B is also connected to A, and traversal is possible in both directions.

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Define a connected graph.
2 marks

Q5 Define a connected graph.

2 marks

Define a connected graph.

Your Answer

A connected graph is a graph in which there is a path between every pair of vertices. This means every vertex can be reached from any other vertex through the edges of the graph. If no such path exists between any pair of vertices, the graph is called disconnected.

Save Answer